

# Operadores de Wirtinger

**Definición 1 (Operadores de Wirtinger).** Sea  $f: \Omega \subset \mathbb{C} \longrightarrow \mathbb{C}$ , con  $\Omega$  un dominio,  $\partial f, \bar{\partial} f: \Omega \longrightarrow \mathbb{C}$  son los operadores de Wirtinger

$$\iff \forall x + yi \in \Omega : \begin{cases} \partial f(x + yi) = \partial_z f(x + yi) = \frac{1}{2} \left( \frac{\partial f}{\partial x}(x, y) - i \frac{\partial f}{\partial y}(x, y) \right) \\ \bar{\partial} f(x + yi) = \partial_{\bar{z}} f(x + yi) = \frac{1}{2} \left( \frac{\partial f}{\partial x}(x, y) + i \frac{\partial f}{\partial y}(x, y) \right) \end{cases}$$

donde  $\partial_x f = u_x + iv_x$  y  $\partial_y f = u_y + iv_y$ .

Estos operadores provienen de aplicar la regla de la cadena. Como  $x = \frac{z + \bar{z}}{2}$  y  $y = \frac{z - \bar{z}}{2i}$

$$\implies \partial_z x = \partial_{\bar{z}} x = \frac{1}{2} \wedge \partial_z y = -i \partial_{\bar{z}} y = \frac{1}{2i}.$$

Por lo tanto,  $\partial f = \partial_x f \partial_z x + \partial_y f \partial_z y = \partial_x f \frac{1}{2} + \partial_y f \frac{1}{2i} = \frac{1}{2} (\partial_x f - i \partial_y f)$  y

$$\bar{\partial} f = \partial_x f \partial_{\bar{z}} x + \partial_y f \partial_{\bar{z}} y = \partial_x f \frac{1}{2} + \partial_y f \frac{-1}{2i} = \frac{1}{2} (\partial_x f + i \partial_y f).$$